Supplementary Material: Phase-dependent recovery and lifetime of encoded entanglement under collective detector noise

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This document gives self-contained proofs and specifies the regulated model. Equation numbers beginning with S refer to this document. We use $P_j = |j\rangle\langle j|$, $|g\rangle = |00\rangle$, $|e\rangle = |11\rangle$, $|t,s\rangle = (|01\rangle \pm |10\rangle)/\sqrt{2}$, $P_T = I - P_s$, and $\mathcal{N} = \|\rho^{T_A}\|_1 - 1$ across $A|BCD$.

S1 Single-detector protection and sector recovery

For $|\mathrm{CL}_4\rangle = (|0000\rangle + |0011\rangle + |1100\rangle - |1111\rangle)/2$,

$$\mathrm{tr}_D\, P_{\mathrm{CL}_4} = \tfrac{1}{2} P^{AB}_{\Phi^+} \otimes P^C_0 + \tfrac{1}{2} P^{AB}_{\Phi^-} \otimes P^C_1, \qquad |\Phi^\pm\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2}. \tag{S1}$$

The orthogonal C flags make the negativity across $A|BC$ equal to one. An arbitrary trace-preserving channel on D leaves this reduced state unchanged. Discarding D cannot increase negativity, while the qubit-reference upper bound is one; the full output therefore has negativity one. Exchange of C, D gives the other case. Equivalently, the logical vectors $|0\rangle_B|\Phi^+\rangle_{CD}$ and $|1\rangle_B|\Phi^-\rangle_{CD}$ have equal reduced diagonal operators $I/2$ and zero reduced off-diagonal operators on either erased detector. These are the erasure form of the Knill–Laflamme conditions [1]. Local preparation unitaries on C, D preserve this property.

The prepared family can be written as

$$\begin{aligned} |\psi_\theta\rangle &= \frac{|00\rangle_{AB}(|01\rangle + e^{i\theta}|10\rangle)_{CD} + |11\rangle_{AB}(|01\rangle - e^{i\theta}|10\rangle)_{CD}}{2} \\ &= \frac{|\phi_T\rangle_{AB}|t\rangle + |\phi_s\rangle_{AB}|s\rangle}{\sqrt{2}}, \\ |\phi_T\rangle &= \tfrac{1}{2}[(1+e^{i\theta})|00\rangle + (1-e^{i\theta})|11\rangle], \\ |\phi_s\rangle &= \tfrac{1}{2}[(1-e^{i\theta})|00\rangle + (1+e^{i\theta})|11\rangle]. \end{aligned} \tag{S2}$$

Both AB vectors are normalized and have negativity $h = |\sin\theta|$. For a completely positive trace-preserving channel $\mathcal{E}(\rho) = \sum_\mu K_\mu \rho K_\mu^\dagger$ with $[K_\mu, P_s] = 0$, a subsequent nonselective sector measurement leaves these AB states with orthogonal detector flags, each of weight $1/2$. Its negativity is h. Monotonicity under this register-side measurement gives $\mathcal{N} \geq h$. This argument uses Kraus-level sector preservation, not merely exchange covariance.

At $\theta = \pi/2$, set $U_T = \mathrm{diag}(1, i)_B$ and $U_s = \mathrm{diag}(1, -i)_B$. Directly,

$$(I_A \otimes U_T)|\phi_T\rangle = e^{i\pi/4}|\Phi^+\rangle, \qquad (I_A \otimes U_s)|\phi_s\rangle = e^{-i\pi/4}|\Phi^+\rangle. \tag{S3}$$

Consequently $V = U_T \otimes P_T + U_s \otimes P_s$ maps the prepared state to $|\Phi^+\rangle_{AB} \otimes |\xi\rangle_{CD}$, where $|\xi\rangle = (e^{i\pi/4}|t\rangle + e^{-i\pi/4}|s\rangle)/\sqrt{2}$. Since $[V, I_B \otimes K_\mu] = 0$, the same operation after the channel gives

$$(I_A \otimes V)(\mathrm{id}_{AB} \otimes \mathcal{E})(P_{\psi_{\pi/2}})(I_A \otimes V^\dagger) = P_{\Phi^+} \otimes \mathcal{E}(P_\xi). \tag{S4}$$

Joint operations within BCD are essential to the stated coherent decoder. This identity also permits decoding before the noise and therefore is not a comparison with unrestricted joint preprocessing.

S2 Regulated field model and established spectra

Let $a > 0$ be the proper acceleration, $L \geq 0$ the transverse separation and τ the central proper time. The detector centres follow

$$x_j(\tau) = (a^{-1}\sinh a\tau, a^{-1}\cosh a\tau, y_j, 0), \qquad y_C = 0, \quad y_D = L. \tag{S5}$$

The interaction is

$$H_I(\tau) = \lambda\chi(\tau)\sum_{j=C,D}(\sigma_j^- e^{-i\omega\tau} + \sigma_j^+ e^{i\omega\tau})\Phi_j(\tau). \tag{S6}$$

Here $\omega > 0$ is the common energy gap, λ is a dimensionless weak coupling, χ is a real smooth switching function, and $\sigma_j^- = |0\rangle_j\langle 1|$ with $\sigma_j^+ = (\sigma_j^-)^\dagger$. The field and register start independently. Both detectors have the same gap, coupling and switching, followed by a constant-coupling stage. With a smooth real profile of support radius $r < a^{-1}$, normalized by $\int F_r\, d^3\xi = 1$, define [2]

$$\Phi_j(\tau) = \int d^3\xi\, F_r(\boldsymbol{\xi})\phi((a^{-1}+\xi_x)\sinh a\tau, (a^{-1}+\xi_x)\cosh a\tau, y_j+\xi_y, \xi_z). \tag{S7}$$

The translated profiles have even transverse dependence and remain inside the right Rindler wedge. Their vacuum correlations are stationary under the boost and obey the Kubo–Martin–Schwinger condition at inverse central proper-time temperature $\beta = 2\pi/a$. Finite r defines the microscopic coupling and its renormalized Hamiltonian. Point-profile formulas below describe dissipative spectral coefficients, not an unregulated Lamb shift.

In that point-profile limit, the known vacuum two-position correlation is [3]

$$G_{ij}^+(\Delta\tau) = -\frac{a^2}{16\pi^2}\left[\sinh^2\frac{a(\Delta\tau - i0)}{2} - \frac{a^2L_{ij}^2}{4}\right]^{-1}, \quad L_{ii} = 0, \quad L_{CD} = L. \tag{S8}$$

Use $\mathcal{G}_{ij}(\nu) = \int d\Delta\tau\, e^{i\nu\Delta\tau}G_{ij}^+(\Delta\tau)$, with positive ν representing de-excitation. Put $b_L = \operatorname{arsinh}(aL/2)$ and $\ell_L = 2b_L/a$. For $L > 0$ the poles are $\Delta\tau = \pm\ell_L + i0 + in\beta$. At $\nu > 0$ close above. The pair of residues in the lowest strip sums to

$$-\frac{i\sin(\nu\ell_L)}{4\pi^2 L\cosh b_L}.$$

Summing the factors $e^{-n\beta\nu}$ and multiplying by $2\pi i$ yields

$$\begin{aligned}\mathcal{G}_{CD}(\nu) &= \frac{\sin(\nu\ell_L)}{2\pi L\cosh b_L(1-e^{-\beta\nu})},\\ \mathcal{G}_{\mathrm{loc}}(\nu) &= \frac{\nu}{2\pi(1-e^{-\beta\nu})}, \qquad f_{\mathrm{acc}} = \frac{\sin[(2\omega/a)\operatorname{arsinh}(aL/2)]}{\omega L\sqrt{1+a^2L^2/4}}.\end{aligned} \tag{S9}$$

Negative frequencies follow from $\mathcal{G}(-\nu) = e^{-\beta\nu}\mathcal{G}(\nu)$. For the inertial thermal field, its momentum integral at $\nu > 0$ gives the on-shell delta function $2\pi\delta(\nu - k)$ and the angular integral $4\pi\,\mathrm{sinc}(\nu L)$. Thus $\mathcal{G}_{CD}^{\mathrm{th}}(\nu) = \mathcal{G}_{\mathrm{loc}}(\nu)\,\mathrm{sinc}(\nu L)$. These spectra and their spatial distinction are established inputs [3].

The spectral matrix is $\mathcal{G}_{\mathrm{loc}}(\nu)\begin{pmatrix}1 & f\\ f & 1\end{pmatrix}$; positivity gives $|f| \leq 1$. Its symmetric and antisymmetric eigenvectors give $J_\pm = (\sigma_C^- \pm \sigma_D^-)/\sqrt{2}$ and rates $(1 \pm f)\gamma_{\downarrow,\uparrow}$. The secular weak-coupling generator retains these collective terms because the detector gaps are degenerate. It is the collective generator specified in the main text, with $\gamma_\downarrow = \lambda^2\mathcal{G}_{\mathrm{loc}}(\omega)$ and $\gamma_\uparrow = q\gamma_\downarrow$, $q = e^{-\beta\omega}$. Define $\Gamma = \gamma_\downarrow + \gamma_\uparrow$ and $z = \Gamma\tau$; $\mathcal{E}_z$ is the detector channel at proper time $\tau = z/\Gamma$. The population equations are those of the established collective-detector model [3]. Only a Hamiltonian diagonal in (g, t, e, s) is used in the trace-norm calculation.

For completeness,

$$\frac{2}{a}\operatorname{arsinh}(aL/2) = L - \frac{a^2L^3}{24} + O(L^5), \qquad (1 + a^2L^2/4)^{-1/2} = 1 - \frac{a^2L^2}{8} + O(L^4). \tag{S10}$$

The sine numerator contributes $1 - (\omega^2/6 + a^2/24)L^2$ after division by ωL. Hence $1 - f_{\mathrm{acc}} = (\omega^2 + a^2)L^2/6 + O(L^4)$, whereas $1 - f_{\mathrm{th}} = \omega^2 L^2/6 + O(L^4)$.

S3 Trace norm, phase optimum and recovered state

Write $|\psi_\theta\rangle = (|00\rangle|\zeta_+\rangle + |11\rangle|\zeta_-\rangle)/\sqrt{2}$, with $|\zeta_\pm\rangle = (|01\rangle \pm e^{i\theta}|10\rangle)/\sqrt{2}$. For $W = \mathcal{E}_z(|\zeta_+\rangle\langle\zeta_-|)$, the evolved state is

$$\begin{aligned}\rho(z) = \tfrac{1}{2}\big[&P_{00} \otimes \mathcal{E}_z(P_{\zeta_+}) + P_{11} \otimes \mathcal{E}_z(P_{\zeta_-}) \\ &+ |00\rangle\langle 11| \otimes W + |11\rangle\langle 00| \otimes W^\dagger\big].\end{aligned} \tag{S11}$$

Partial transpose on A moves the cross terms to the $AB = 01, 10$ block, which is $\frac{1}{2}\begin{pmatrix} 0 & W^\dagger \\ W & 0 \end{pmatrix}$. Its eigenvalues are $\pm\varsigma_j(W)/2$, where ς_j are singular values. The other two blocks are positive, each of trace $1/2$. Therefore $\|\rho^{T_A}\|_1 = 1 + \|W\|_1$ and $\mathcal{N} = \|W\|_1$.

Put $\alpha = \sin\theta/2$, $b = (1 + \cos\theta)/2$, $c = (1 - \cos\theta)/2$; then $bc = \alpha^2$ and

$$|\zeta_+\rangle\langle\zeta_-| = i\alpha(P_t - P_s) + b|t\rangle\langle s| + c|s\rangle\langle t|. \tag{S12}$$

Let $\boldsymbol{d} = e^{Mz}(0, 1, 0, -1)^{\mathsf{T}}$, with M given below. The t–s coherence has decay e^{-z} and phase $\varphi = (E_t - E_s)z/\Gamma$, where E_t, E_s are the corresponding Hamiltonian eigenenergies, so W has scalar entries $i\alpha d_g, i\alpha d_e$ and the block

$$Y = \begin{pmatrix} i\alpha d_t & be^{-z-i\varphi} \\ ce^{-z+i\varphi} & i\alpha d_s \end{pmatrix}. \tag{S13}$$

The identity $\|Y\|_1^2 = \operatorname{tr}(YY^\dagger) + 2|\det Y|$ follows by adding its two singular values. Here

$$\operatorname{tr}(YY^\dagger) = \alpha^2(d_t^2 + d_s^2) + (b^2 + c^2)e^{-2z}, \qquad \det Y = -\alpha^2(d_t d_s + e^{-2z}). \tag{S14}$$

The Hamiltonian phase cancels. With $h = |\sin\theta|$ and $E = e^{-2z}$ this proves

$$\mathcal{N}_\theta = \frac{h}{2}(|d_g| + |d_e|) + \sqrt{E + h^2 K}, \quad K = \frac{d_t^2 + d_s^2 - 2E + 2|d_t d_s + E|}{4}. \tag{S15}$$

If $d_t d_s + E \geq 0$, then $K = (d_t + d_s)^2/4$. Otherwise $d_t d_s < -E$ and $(d_t - d_s)^2 \geq -4d_t d_s > 4E$, giving $K \geq 0$ again. Monotonicity in h establishes the phase optimum.

A projective Dicke measurement removes the off-diagonal detector coherences. The unnormalized AB state for outcome j has off-diagonal entry $i\alpha d_j/2$. A phase gate on B makes it real and nonnegative. Since all outcomes are kept, the resulting off-diagonal entry is $h\sum_j |d_j|/4$. Both diagonal entries sum to $1/2$, giving

$$\rho_{AB}^{\mathrm{rec}} = \frac{1 + N_r}{2}P_{\Phi^+} + \frac{1 - N_r}{2}P_{\Phi^-}, \qquad N_r = \frac{h}{2}\sum_j |d_j|. \tag{S16}$$

Register-side monotonicity gives $N_r \leq \mathcal{N}_\theta$. The procedure is fixed to this measurement and conditional correction; no general recovery optimum is asserted. Standard teleportation using this Bell mixture is the dephasing channel $\rho \mapsto (1 + N_r)\rho/2 + (1 - N_r)\sigma_z\rho\sigma_z/2$. Its Bloch-sphere average fidelity is $(2 + N_r)/3$, consistently with the standard singlet-fraction relation [4].

At $q = 0$, the exact population difference is

$$d_t = e^{-(1+f)z}, \quad d_s = -e^{-(1-f)z}, \quad d_e = 0, \quad d_g = e^{-(1-f)z} - e^{-(1+f)z}. \tag{S17}$$

Consequently

$$\mathcal{N}_\theta = e^{-z}\left[h|\sinh(fz)| + \sqrt{1 + h^2\sinh^2(fz)}\right], \qquad \mathcal{N}_{\pi/2} = e^{-(1-|f|)z}. \tag{S18}$$

This treats the zero-temperature boundary separately from the fixed-$q > 0$ projection.

S4 Slow-sector projection and contrast

For a population column vector ordered as (g, t, e, s), define

$$M = \begin{pmatrix} -q(r_+ + r_-) & r_+ & 0 & r_- \\ qr_+ & -(1+q)r_+ & r_+ & 0 \\ 0 & qr_+ & -(r_+ + r_-) & qr_- \\ qr_- & 0 & r_- & -(1+q)r_- \end{pmatrix}, \qquad r_\pm = \frac{1 \pm f}{1+q}. \tag{S19}$$

At $f = 1 - \epsilon$, $M = M_0 + \epsilon M_1$ exactly, where

$$M_1 = \frac{1}{1+q}\begin{pmatrix} 0 & -1 & 0 & 1 \\ -q & 1+q & -1 & 0 \\ 0 & -q & 0 & q \\ q & 0 & 1 & -(1+q) \end{pmatrix}. \tag{S20}$$

The right zero vectors of M_0 are $\boldsymbol{r}_T = (1, q, q^2, 0)^\mathsf{T}/Z$ and $\boldsymbol{r}_s = (0, 0, 0, 1)^\mathsf{T}$, with $Z = 1 + q + q^2$. The left vectors $\boldsymbol{\ell}_T = (1, 1, 1, 0)$ and $\boldsymbol{\ell}_s = (0, 0, 0, 1)$ obey $\boldsymbol{\ell}_i \boldsymbol{r}_j = \delta_{ij}$. The fast population eigenvalues are $-2 \pm 2\sqrt{q}/(1+q)$ and stay away from zero. Projection gives

$$[\boldsymbol{\ell}_i M_1 \boldsymbol{r}_j]_{i,j=T,s} = \begin{pmatrix} -q/Z & 1 \\ q/Z & -1 \end{pmatrix}. \tag{S21}$$

Its difference eigenvector $(1, -1)^\mathsf{T}$ has eigenvalue $-\mathcal{C}(q) = -(1 + q/Z)$. The initial $\boldsymbol{d}(0)$ projects to this difference and has no stationary component. Finite-dimensional perturbation theory then gives

$$e^{Mu/\epsilon}(0, 1, 0, -1)^\mathsf{T} \longrightarrow e^{-\mathcal{C}(q)u}(\boldsymbol{r}_T - \boldsymbol{r}_s), \qquad u > 0. \tag{S22}$$

The two limiting population vectors have disjoint supports and each sums to one. The t–s coherence vanishes as $e^{-u/\epsilon}$, so both Eq. (S15) and Eq. (S16) tend to $he^{-\mathcal{C}(q)u}$. The physical decay rate is $\kappa = \epsilon\Gamma\mathcal{C}(q) + O(\epsilon^2\Gamma)$. At fixed $q > 0$, errors are $O(\epsilon)$ on compact positive intervals of u, together with fast exponentially decaying terms. The fast boundary layer prevents a uniform assertion at $u = 0$ for general θ.

The short-time expansion is distinct: $d_t = 1 - (1+f)z + O(z^2)$, $d_s = -1 + (1-f)z + O(z^2)$, $d_g = 2fz/(1+q) + O(z^2)$ and $d_e = 2qfz/(1+q) + O(z^2)$. For $f \geq 0$, substitution gives $\mathcal{N}_{\pi/2} = 1 - (1-f)z + O(z^2)$.

For matched local rates and $R = 1 + a^2/\omega^2 > 1$, the limiting resource difference is $\Delta(w) = e^{-\mathcal{C}w} - e^{-\mathcal{C}Rw}$. Its derivative vanishes when $e^{\mathcal{C}(R-1)w} = R$. This is the unique positive maximum, yielding

$$w_* = \frac{\ln R}{\mathcal{C}(R-1)}, \qquad \Delta(w_*) = (1 - R^{-1})R^{-1/(R-1)}. \tag{S23}$$

It vanishes continuously as $R \to 1$. The finite-L generator and the limiting formulas are compared using the supplied numerical scripts; the optimization concerns this resource difference.

S5 Common-coupling symmetry and slow-time stability

At $L = 0$, Eq. (S7) gives $\Phi_C = \Phi_D$. The identical-gap free detector Hamiltonian and the interaction commute with P_s. Hence the full time-ordered unitary commutes with P_s; its bath matrix elements yield sector-preserving Kraus operators for any initially independent bath state. The protection in Section S1 is exact under these assumptions.

At finite profile radius, a transverse translation contributes e^{ik_yL} to the cross spectrum. Reflection symmetry removes the imaginary part. With a positive normalized on-shell spectral measure $d\mu_r$ and finite fourth moment,

$$f_r(L) = \int \cos(k_yL)\, d\mu_r = 1 - c_rL^2 + O(L^4), \qquad c_r = \tfrac{1}{2}\int k_y^2\, d\mu_r. \tag{S24}$$

For an approximate-identity profile family whose on-shell second moments converge, $c_r^{\mathrm{acc}} \to (\omega^2 + a^2)/6$ and $c_r^{\mathrm{th}} \to \omega^2/6$. The local regulated rates may be matched by calibrating coupling strengths. The point-profile ratio follows only with these moment and matching assumptions.

At each fixed $r, L > 0$, assume convergence to the weak-coupling generator uniformly on bounded intervals of scaled time $z = \Gamma_{\lambda,r}\tau$ [5]. For fixed L and fixed $u > 0$, $z = u/[1 - f_r(L)]$ is finite. Thus taking $\lambda \to 0$ before $L \to 0$ gives the conditional ordered microscopic limit stated in the main text. It is not a simultaneous-limit theorem, nor does it bound memory errors at fixed coupling on all growing times. The scale restrictions $\Gamma \ll \omega$ and $\Gamma\tau_B \ll 1$, including Hamiltonian compatibility, remain necessary for the effective description [6, 7].

Finally let both $\mathscr{L}, \widetilde{\mathscr{L}}$ generate time-independent completely positive trace-preserving evolutions, with $\|\widetilde{\mathscr{L}} - \mathscr{L}\|_\diamond \le \delta\Gamma$. Duhamel's identity is

$$e^{\tau\widetilde{\mathscr{L}}} - e^{\tau\mathscr{L}} = \int_0^\tau dv\, e^{(\tau-v)\widetilde{\mathscr{L}}}(\widetilde{\mathscr{L}} - \mathscr{L})e^{v\mathscr{L}}. \tag{S25}$$

Each channel has diamond norm one, so the channel difference is bounded by $\delta\Gamma\tau$. For states with a qubit reference,

$$|\mathcal{N}(\widetilde{\rho}) - \mathcal{N}(\rho)| \le \|(\widetilde{\rho} - \rho)^{T_A}\|_1 \le 2\|\widetilde{\rho} - \rho\|_1. \tag{S26}$$

It follows that $|\widetilde{\mathcal{N}} - \mathcal{N}| \le 2\delta\Gamma\tau = 2\delta u/\epsilon$. Requiring $\delta/\epsilon \to 0$ is a sufficient way to retain the slow law. For a near-degenerate collective description, a Hamiltonian gap mismatch contributes a generator perturbation of order $|\Delta\omega|$; the sufficient comparison is $|\Delta\omega| \ll \epsilon\Gamma$. The bound need not be tight for a specified perturbation.

S6 Numerical verification and its scope

The accompanying archive reruns the original supplied finite-dimensional model independently of its pre-existing output files. Full four-qubit evolution was compared with the trace-norm expression for 800 parameter cases, including two exchange-Hamiltonian values. The maximum absolute disagreement was 2.13×10^{-15}. Two thousand phase comparisons found no violation above 2.23×10^{-16}. Random local channels, nonunitary sector-preserving channels, the decoder, recovered state and teleportation channel were also checked.

The revision adds `src/validate_submission.py`. It checks the compact trace-norm formula against full evolution for 300 cases on $q \in \{0, 0.01, 0.4, 0.95\}$, $f \in \{-1, 0, 0.6, 0.999, 1\}$, $\theta \in \{0, 0.37, \pi/2\}$ and $z \in \{0, 0.01, 1, 12, 500\}$. The largest absolute difference is 2.60×10^{-13}, including the long-time cases. These are floating-point comparisons, not exact symbolic equalities.

For slow convergence the grids are

$$q \in \{0.01, 0.1, 0.4, 0.9\},$$
$$\epsilon \in \{0.1, 0.05, 0.02, 0.01, 0.005, 0.002\}.$$

Eigenvalues give 24 rate comparisons. Taking $\theta \in \{0.37, \pi/2\}$ and $u \in \{0.5, 1, 2\}$ gives 144 comparisons of both the full and recovered negativity with the leading law. For each sampled parameter family the endpoint errors decrease as ϵ decreases. At $\epsilon = 0.002$, the largest relative rate error is 1.26×10^{-4} and the full-negativity error is 1.52×10^{-4}. The positive lower limit on u avoids claiming uniform convergence through the initial layer.

Contrast is tested for $a/\omega \in \{0.5, 1, 2, 3\}$ and $\omega L \in \{0.4, 0.2, 0.1, 0.05\}$. Each case first uses a 501-point logarithmic grid over $w/w_* \in [10^{-5}, 100]$ and then bounded continuous optimization around its largest sampled value. The optimized time and contrast approach their leading formulas as the separation decreases; at $\omega L = 0.05$ the largest sampled contrast error is 9.22×10^{-4}. This numerical search checks a broad range of times and supports the limiting calculation, without proving global finite-separation optimality for every parameter.

The corresponding files are `submission_slow_rates.csv`, `submission_slow_law.csv`, `submission_contrast.csv` and `submission_validation.json`. The original plot samples and error tables are also supplied.

This verification concerns the given channel model on a finite Hilbert space. It does not prove novelty, establish experimental feasibility, or validate a uniform microscopic long-time approximation. Some channel tests use negative spatial factors within the mathematical range of complete positivity. The environmental plots use the stated physical spectra.

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